Project Plan

Draft Title: Graph-Based Clustering and Classification of Publicly Labelled Ethereum Addresses Using Public On-Chain Data

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1. Short Project Summary

This project investigates whether public Ethereum on-chain data can distinguish address categories (CEX, DeFi, Malicious_MEV) through graph-based analysis. Distinguishing these categories matters for on-chain risk monitoring and compliance analytics, as different entity types exhibit different transaction patterns that blockchain security practitioners need to identify at scale. Focusing on address-level classification rather than real-world deanonymisation, I will construct a 90-day sampled transaction ego-network around publicly labelled seeds. By extracting transaction behaviours, graph structural features, and Node2Vec embeddings, I will evaluate baseline machine learning and clustering methods. The expected outcome is a reproducible technical pipeline and an empirical evaluation of the effectiveness of these graph signals in classifying blockchain address categories.

2. Research Question(s) or Objectives

1. To what extent can transaction behaviour features and graph structural features distinguish publicly labelled Ethereum address categories?
2. Do node embedding methods capture meaningful structural patterns in the sampled Ethereum transaction graph, and do they improve clustering or classification compared with hand-crafted features?

3. Background and Motivation

Ethereum transactions naturally form a directed graph where addresses are nodes and transfers are edges. Consequently, an address is defined not only by isolated transaction statistics but also by its counterparties and topological position. This structural complexity motivates the use of graph-based methods for blockchain analytics, including node embedding approaches such as Node2Vec (Grover and Leskovec, 2016) and graph neural networks such as GraphSAGE (Hamilton, Ying and Leskovec, 2017).

This project aims to distinguish different blockchain entities—specifically CEX wallets, DeFi contracts, and Malicious_MEV actors—using public on-chain data. While public labels offer a

valuable supervision signal, Ethereum data presents significant practical challenges: labels are noisy, activity distributions are highly heavy-tailed (with extreme outliers and zero-value transactions), and the full network is computationally prohibitive. Prior work on Ethereum address classification has shown that graph structural features can improve detection of phishing and fraudulent accounts (Chen et al., 2020), but most studies rely on curated datasets that may not reflect these real-world data constraints.

Motivated by these realistic constraints, this dissertation focuses on constructing a feasible, 90-day sampled transaction ego-network to explore behavioral and topological differences between address categories. It evaluates how effectively engineered graph features and embeddings can differentiate these entities. Crucially, the project relies entirely on public ledgers and does not attempt to deanonymise real-world users or attribute legal criminality; it treats labels strictly as functional address categories for technical evaluation.

4. Data Source(s)

The project uses publicly available Ethereum on-chain data and public address label sources. The project utilizes public address labels initially sourced from Dune Analytics, alongside on-chain transaction data from Google Cloud BigQuery (crypto_ethereum.transactions).

The initial raw seeds included 300 MEV, 150 DeFi, and 150 CEX addresses. After removing duplicates (e.g., merging MEV down to 299) and using BigQuery to filter for addresses active within the last 90 days, the final cleaned dataset contains 500 unique seed nodes: Malicious_MEV (299), DeFi (150), and CEX (51).

Using a top-100 counterparty sampling strategy, I constructed a structurally clean directed ego-network graph comprising 12,861 aggregated edges and 8,029 unique nodes.

Known Data Issues and Scope Limitations:

1. Sampling Bias: The top-100 limit provides a computationally feasible but partial network view.
2. DeFi Under-representation: Only 57.3% (86/150) of DeFi seeds are active in the direct ETH transaction graph, as much of their activity occurs in DEX logs or internal traces.
3. Zero-Value Edges: Intentionally retained, as they frequently represent smart contract calls.
4. Label Noise: Public labels may be incomplete or outdated.

5. Proposed Methodology

The project follows a graph-based empirical pipeline. First, I clean the public labelled seed addresses and construct a 90-day directed Ethereum transaction ego-network around the final 500 seeds. To keep the graph computationally feasible, I retain the top 100 counterparties per

seed and aggregate repeated transactions into weighted directed edges. Basic sanity checks have confirmed no missing source or target addresses, no self-loops, and no duplicated source-target edges.

Next, I will engineer address-level transaction behaviour features, including incoming and outgoing transaction counts, ETH value, gas fees, zero-value transaction ratios, and log-transformed versions of skewed variables. I will also compute graph structural features such as in-degree, out-degree, weighted degree, PageRank, connected component size, and clustering coefficients.

I will then apply Node2Vec to learn graph embeddings for visualisation, clustering, and classification. For clustering, I will use K-Means and Louvain community detection, evaluated using NMI, ARI, and Silhouette Score, to assess whether the learned embeddings naturally separate address categories. For supervised training, because the labelled classes are imbalanced, I will use a balanced labelled subset of approximately 50 seeds per class where appropriate, while also reporting results on the full dataset. Baseline classifiers such as Logistic Regression, Random Forest, and possibly XGBoost will be evaluated using macro-F1, balanced accuracy, and class-level precision and recall. To understand the relative contribution of different feature types, I will conduct a feature group ablation study comparing transaction behaviour features alone, graph structural features alone, and the combined feature set. I will also apply SHAP and feature importance analysis to interpret which specific features drive model predictions across address categories. If time allows, GraphSAGE may be added as an optional extension.

6. Ethics Route

This project falls under Category 01 (no ethical approval required). It uses publicly available Ethereum blockchain transaction data and public address labels. No data is collected from human participants or animals. The project does not involve surveys, interviews, or experimental interventions. It does not scrape forums or social media, and it does not attempt to re-identify individuals behind blockchain addresses. Labels are used strictly as public address-level category labels for technical analysis.

7. Expected Contribution or Possible Outcome

The expected contribution is a reproducible public-data pipeline for graph-based analysis of labelled Ethereum addresses. The project will produce a cleaned seed dataset, a sampled directed transaction graph, address-level transaction and structural features, Node2Vec embeddings, clustering results, and baseline classification results. These outputs will be used to examine whether public address labels align with observable graph-based behaviour and whether embedding-based representations add value beyond hand-crafted features.

The project also aims to provide an interpretable analysis of which transaction and graph signals are most useful for separating address categories. Its contribution is not to build a production

risk system or identify real-world users, but to evaluate the usefulness and limitations of public on-chain graph data for address-level category analysis.

8. Key Milestones and Timing

Stages 1–4 (project scoping, seed collection, graph construction, and sanity checks) are already complete as of late June 2026. The remaining work is scheduled week by week below, targeting the thesis submission deadline of 4 September 2026.

week	Dates	Tasks
1–4	Before 25 Jun	Project scoping, seed address collection (500 seeds), 90-day directed graph construction (12,861 edges, 8,029 nodes), sanity checks.
5	25 Jun – 1 Jul	Engineer transaction behaviour features (counts, value, gas, ratios, log transforms). Compute graph structural features (degree, PageRank, clustering coefficient, component size).
6	2 Jul – 8 Jul	Finalise feature matrix. Exploratory analysis of feature distributions by class. Handle zero-inflation and skewness.
7	9 Jul – 15 Jul	Train Node2Vec embeddings. Produce UMAP/t-SNE visualisations. Run K-Means and Louvain clustering; evaluate with NMI, ARI, Silhouette Score.
8	16 Jul – 22 Jul	Train baseline classifiers (Logistic Regression, Random Forest, XGBoost) with class weighting and stratified splits. Report macro-F1, balanced accuracy, class-level precision/recall.
9	23 Jul – 29 Jul	Feature group ablation (transaction features only, graph features only, combined). SHAP and feature importance analysis.
10	30 Jul – 5 Aug	Robustness and imbalance checks (balanced subset test, sensitivity analysis). Optional: attempt GraphSAGE if core pipeline is complete.
11	6 Aug – 12 Aug	Write Introduction, Background, Data and Graph Construction, Methodology. Prepare pipeline diagram and dataset summary table.
12	13 Aug – 19 Aug	Write Experiments and Results. Produce all figures (confusion matrices, SHAP plots, embedding plots) and tables.

13	20 Aug – 26 Aug	Write Discussion, Limitations, Ethics, Future Work, Conclusion. Complete appendices. First full draft review and revision.
14	27 Aug – 4 Sep	Final revision, proofreading, formatting checks, and submission buffer.

9. Main Risks and Mitigation

Risk 1: DeFi under-representation
Only 86 of 150 DeFi seeds appear active due to mismatch between DEX-based selection and direct transaction edges.
Mitigation: Report as limitation, retain inactive seeds with proper handling, run active-only check, avoid overclaiming.

Risk 2: Class imbalance
Dataset skewed (299 Malicious_MEV, 150 DeFi, 51 CEX).
Mitigation: Use stratified splits, class weights, macro-F1, balanced accuracy, and a balanced subset test.

Risk 3: Heavy-tailed activity
Few seeds dominate transactions and ETH value.
Mitigation: Apply log1p transforms and report skew as inherent property.

Risk 4: Label noise
Public labels may be incomplete or outdated.
Mitigation: Frame as category classification, avoid identity claims, discuss limitations.

Risk 5: Sampling bias
Top-100 counterparty sampling may omit weaker links.
Mitigation: Clarify method, treat as sampled graph, test sensitivity if possible.

Risk 6: Scope risk
Too many methods may reduce quality.
Mitigation: Prioritise core pipeline; keep GraphSAGE optional.

10. References

The final reference list will be formatted consistently. Initial references may include:

Chen, W., Guo, X., Chen, Z., Zheng, Z. and Lu, Y. (2020) 'Phishing Scam Detection on Ethereum: Towards Financial Security for Blockchain Ecosystem', Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI), pp. 4506–4512.

Grover, A. and Leskovec, J. (2016) 'node2vec: Scalable Feature Learning for Networks', Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 855–864.

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Kipf, T.N. and Welling, M. (2017) 'Semi-Supervised Classification with Graph Convolutional Networks', Proceedings of the 5th International Conference on Learning Representations (ICLR).

Lundberg, S.M. and Lee, S.-I. (2017) 'A Unified Approach to Interpreting Model Predictions', Advances in Neural Information Processing Systems, 30, pp. 4765–4774.

Perozzi, B., Al-Rfou, R. and Skiena, S. (2014) 'DeepWalk: Online Learning of Social Representations', Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 701–710.

Poursafaei, F., Hamad, G.B. and Zilic, Z. (2020) 'Detecting Malicious Ethereum Entities via Application of Machine Learning Classification', 2020 2nd Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS), pp. 120–127. IEEE.

Weber, M., Domeniconi, G., Chen, J., Weidele, D.K.I., Bellei, C., Robinson, T. and Leiserson, C.E. (2019) 'Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics', KDD Workshop on Anomaly Detection in Finance.

Google Cloud (no date) Ethereum Blockchain Dataset on BigQuery. Available at: https://cloud.google.com/blog/products/data-analytics/ethereum-bigquery-public-dataset-smart-contract-analytics (Accessed: June 2026).

Dune Analytics (no date) Public query platform and community datasets. Available at: https://dune.com (Accessed: June 2026).

Ethereum Foundation (no date) Ethereum Documentation: Transactions, Accounts, Gas, and Smart Contracts. Available at: https://ethereum.org/en/developers/docs (Accessed: June 2026).

Additional blockchain-specific academic references will be added during the background reading stage.